Minimum Wages and Employment

An Empirical Replication of Card & Krueger (1994) with a Within-State Treatment-Intensity Extension

1. Research Question and Motivation

1.1 Research question

Did the increase in New Jersey's state minimum wage — from $4.25 to $5.05 per hour on 1 April 1992 — reduce employment in the fast-food industry?

1.2 Motivation

The canonical competitive labour-market model predicts that a binding minimum wage lowers employment. Card and Krueger (1994) exploited the New Jersey minimum-wage hike as a natural experiment and famously found no evidence of employment loss: in a difference-in-differences (DiD) comparison with neighbouring eastern Pennsylvania, where the minimum wage remained at $4.25, employment in New Jersey actually rose relative to Pennsylvania. The result is among the most influential — and most scrutinised — findings in empirical labour economics, because it calls into question the textbook prediction.

This report has two goals. First, it reproduces the paper's core tables (Tables 2, 3 and 4 and Figure 1) from the underlying store-level survey data, and compares the reproduced estimates with the published values. Second, it provides an independent extension: a within-New-Jersey treatment-intensity DiD that does not rely on Pennsylvania as the control group.

2. Data Source, Sample Construction, and Variable Definitions

2.1 Data source

The analysis uses the Card–Krueger fast-food restaurant survey (public.csv, 410 stores × 46 columns), administered by telephone in two waves: wave 1 in February–March 1992 (before the wage change) and wave 2 in November–December 1992 (after). The sample covers four chains — Burger King, KFC, Roy Rogers and Wendy's — located in New Jersey and eastern Pennsylvania.

2.2 Sample construction

The treatment group consists of the 331 New Jersey stores (STATE = 1); the control group consists of the 79 eastern-Pennsylvania stores (STATE = 0). Employment is measured in full-time equivalents (FTE), defined as in the paper (p. 775):

FTE = full-time employees + managers/assistant managers + 0.5 × part-time employees.

If any of the three components is missing, the store's FTE is recorded as missing (rather than treating the missing component as zero). In wave 2, the six permanently-closed stores (STATUS2 = 3) are assigned zero employment, and the four temporarily-closed stores (STATUS2 $\in$ {2,4,5}) are treated as missing (set to zero in one robustness row). The "balanced panel" uses stores with non-missing FTE in both waves (309 NJ and 75 PA stores).

2.3 Variable definitions

Variable	Definition
STATE	1 = New Jersey (treatment); 0 = Pennsylvania (control)
FTE	EMPFT + NMGRS + 0.5 × EMPPT (full-time + managers + 0.5 × part-time)
ΔFTE	FTE (wave 2) − FTE (wave 1): change in employment
WAGE_ST	Starting wage in wave 1, dollars per hour
GAP	max(0, (5.05 − WAGE_ST) / WAGE_ST): proportional wage rise needed to reach $5.05 (NJ only)
STATUS2	Wave-2 store status (3 = permanent closure → FTE2 = 0; 2/4/5 = temporary closure → missing)

3. Empirical Design

3.1 Difference-in-differences

The DiD estimator compares the before-after change in New Jersey with the before-after change in Pennsylvania:

DiD = (NJ_after − NJ_before) − (PA_after − PA_before).

The first difference removes time-invariant, state-specific characteristics; the second removes shocks that are common to both states. The identifying assumption is parallel trends: in the absence of the minimum-wage increase, employment in New Jersey and Pennsylvania would have followed the same path. Two facts support this assumption: the two areas are adjacent and draw on the same fast-food labour market, and their pre-policy starting wages are nearly identical ($4.61 in NJ vs $4.63 in PA).

3.2 Two complementary identification strategies

- Between-state DiD (NJ vs PA): the primary design.
- Within-NJ treatment-intensity design: because the whole state was treated, there is no 0/1 control group inside New Jersey. Instead, stores were exposed with different intensities: a store with a lower pre-policy starting wage needed a larger

proportional wage increase (GAP) to reach $5.05. Stores already paying ≥ $5.05 have GAP = 0 and serve as the within-state reference group. If the minimum wage reduced employment, stores with larger GAP should have shed more workers.

3.3 Regression specifications

With two periods, the DiD can be estimated by regressing the change in employment on the New Jersey dummy. Table 4 (models i–ii) reports $\Delta FTE_i = \alpha + \beta \cdot NJ_i +$ controls. Models iii–v replace the NJ dummy with the continuous exposure GAP_i (treatment-intensity form). All regressions are estimated by OLS.

4. Core Reproduction Results and Comparison with the Original

4.1 Table 3 — average FTE employment per store

Means with standard errors in parentheses.

	Pennsylvania	New Jersey	Difference (NJ – PA)
FTE before	23.33 (1.35)	20.44 (0.51)	–2.89 (1.44)
FTE after	21.17 (0.94)	21.03 (0.52)	–0.14 (1.07)
Change	–2.16 (1.25)	+0.59 (0.54)	+2.76 (1.36)

Within New Jersey, classified by the wave-1 starting wage, employment rose among the most exposed stores and fell among the least exposed:

Wave-1 wage group	N	Change in FTE
= $4.25	101	+1.32
$4.26–$4.99	140	+0.87
≥ $5.00	73	–2.04
Low – High		+3.36

4.2 Table 2 — means of key variables

The reproduced sample means (chain shares, company ownership, FTE, starting wages, meal prices, hours open, recruiting bonus) match the published values to two decimals. The one exception is the "% full-time employees" row, where the public-use file records a different full-time/part-time mix in wave 1 (≈25% vs the paper's 32.8%); total FTE is unaffected and still matches exactly.

4.3 Table 4 — reduced-form regressions

Dependent variable: change in FTE employment. Coefficient (standard error).

Model	NJ dummy	Wage gap (GAP)	Controls
(i)	+2.36 (1.15)	—	none
(ii)	+2.33 (1.15)	—	chain, ownership

(iii)	—	+15.89 (5.95)	none
(iv)	—	+15.19 (6.08)	chain, ownership
(v)	—	+12.03 (7.32)	chain, ownership, region

4.4 Comparison with the original paper

The reproduced DiD point estimate (+2.76 FTE per store) is identical to the published value, and the levels of FTE before and after match exactly (20.44, 21.03, 23.33, 21.17). Three small differences arise from the public-use data file rather than from the method: (i) the row-3 standard error of the DiD is 1.31 versus 1.36; (ii) the "% full-time" row differs as noted above; and (iii) the regression sample contains 365 stores versus 357 in the paper. None of these affects the point estimates or the conclusion.

5. Independent Extension: Within-NJ Treatment-Intensity DiD

The extension drops every Pennsylvania observation and uses only the variation in exposure within New Jersey. The continuous treatment intensity is GAP_i = max(0, (5.05 – WAGE_ST_i) / WAGE_ST_i). Regressions of ΔFTE on GAP give:

Specification	**GAP coefficient**	**s.e.**	**t**	**p**
GAP only	+13.08	6.60	1.98	0.048
+ chain & ownership	+12.27	6.82	1.80	0.073
+ region	+12.14	6.87	1.77	0.078

The discrete Low–High contrast (wage = $4.25 vs ≥ $5.00) is +3.36 FTE (s.e. 1.28, p = 0.009); the Mid–High contrast is +2.87 (s.e. 1.20, p = 0.017). The positive GAP coefficient means that stores which required a larger wage increase to comply with the new minimum — i.e. the most exposed stores — gained employment relative to the least-exposed stores. This within-state result does not rely on Pennsylvania as a control and points in the same direction as the between-state DiD.

6. Validity Checks, Limitations, and Conclusion

6.1 Validity checks

- Parallel-trends support: adjacent states, near-identical pre-policy starting wages, and a confirmed "bite" (NJ starting wages rose from $4.61 to $5.08 after the change, with 85% of stores at $5.05).
- Robustness of Table 3: the balanced-panel DiD (+2.75) and the temporarily-closed-set-to-zero DiD (+2.51) are close to the headline estimate.

- Within-NJ wage-gradient check: the Low–High contrast (+3.36) is positive and significant, consistent with the between-state result.

6.2 Limitations

- With only two time periods, the parallel-trends assumption cannot be tested directly (no pre-trends are observable).
- The sample is modest (≈300–410 stores), and standard errors are conventional OLS rather than cluster-robust.
- The public-use data file differs slightly from the paper's original wave-1 staffing mix, as noted in Section 4.4.
- The within-NJ design relies on the assumption that, absent the policy, employment changes would be unrelated to GAP.

6.3 Conclusion

The reproduction confirms Card and Krueger's headline result: the DiD estimate of +2.76 FTE workers per store ($t \approx 2.0$) provides no evidence that the New Jersey minimum-wage increase reduced fast-food employment, and the within-state treatment-intensity extension reinforces this conclusion using only New Jersey data. Both the between-state and within-state designs are consistent with a zero or mildly positive employment effect.

References

Card, D., & Krueger, A. B. (1994). Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania. American Economic Review, 84(4), 772–793.